Ogbonna Arinze

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SUMMARY

Senior Fullstack Engineer with 8 years of experience architecting high-performance, scalable web applications using Node.js, TypeScript, React, and Vue.js. Proven track record of taking end-to-end ownership of complex technical projects, from conception to rollout, particularly in healthcare and highly sensitive data environments. Deep expertise in building robust backend systems with PostgreSQL, Redis, and REST APIs, alongside crafting intuitive, high-velocity frontend experiences. Passionate about workflow automation, data analytics, and driving engineering excellence in fast-paced, product-focused teams.

CORE COMPETENCIES

- **Fullstack Architecture:** End-to-end project ownership, scalable system design, monorepos (Nx, Turborepo), and micro-frontend patterns.
- **Backend & APIs:** Node.js, NestJS, TypeScript, REST APIs, PostgreSQL, Prisma, Redis, and JWT authentication with role-based access control (RBAC).
- **Frontend Development:** React, Vue.js, Next.js, modern JavaScript (ESNext), Redux, React Query, Tailwind CSS, and Radix UI.
- **Data & Analytics:** Handling highly sensitive data (healthcare clinical trials), data visualization (Recharts, react-chartjs), and workflow automation.
- **Testing & Quality:** Jest, Playwright, Cypress, ESLint, Prettier, Husky, and CI-driven QA.
- **Cloud & DevOps:** Docker, GitHub Actions, AWS, Vercel, CI/CD pipelines, and automated deployment strategies.
- **Performance & Observability:** Lighthouse, React Profiler, code splitting, bundle optimization, and caching mechanisms.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)
Frontend Engineer — *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a modular **React** Chrome extension, optimizing workflows for clinical site integrations and handling **highly sensitive healthcare data**.
- Maintained and extended the existing **Vue.js** codebase during the migration period, fixing bugs and shipping features in parallel with the React rebuild.
- Co-developed a full-stack **Next.js** chat widget powered by a RAG system using **OpenAI GPT models**, **LangChain**, and **Chroma** vector database, enabling users to query their data in real time via a conversational interface.
- Integrated **GraphQL** and **REST API** endpoints for seamless data fetching, and implemented **JWT-based authentication** with **role-based access control** using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Optimized application performance by implementing lazy loading, code splitting, and dynamic imports, reducing load times significantly as measured via **Lighthouse** and **Chrome DevTools**.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline** via **GitHub Actions**, automating reporting with Allure Reports.

Descasio — Lagos, Nigeria (Hybrid)
Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with software architects to create a scalable frontend architecture that accelerated development by **30%**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application, significantly boosting speed and user experience.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Enhanced development workflow by introducing **Jest** and **Enzyme** automated testing frameworks, leading to a **20% decrease** in bug reports and improved release confidence.
- Established engineering standards for code quality and deployment using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.

ActivEdge Technologies — Lagos, Nigeria
Senior React Developer — *Oct 2019 - Sep 2021*

- Built a high-volume bank Settlement Solution in a team of 3 frontend engineers, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules using **React** with **Redux** for state management, **RxJS** for reactive data streaming, and **Axios** for asynchronous API requests.

- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Built dynamic data visualization dashboards using **React** and **react-chartjs**, streamlining management of user roles and permissions with **Redux Toolkit** and Ant Design UI.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, focusing on testable, maintainable code.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to Heroku and AWS.
- Mentored two interns through frontend and backend projects, providing technical guidance and professional development support.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed SureEdu, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records, and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected and built the entire platform end-to-end as an **Nx monorepo** managing 6 applications, including a **NestJS** API backend, **Next.js** citizen-facing web app, admin dashboard, and document ingestion pipeline.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including the pgvector extension for vector storage, and built a hybrid RAG pipeline.
- Built the document ingestion pipeline supporting PDF, XLSX, DOCX, JSON, and Markdown extractors, processing 700+ documents across 37 Nigerian states with zero ingestion errors.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** end-to-end tests, and **Docker Compose** production deployment with multi-stage Dockerfiles, memory limits, and health checks.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*